

# APM1220 FA 2

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## Problem 3.1

Given

$$X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}.$$

(a)

The three points are

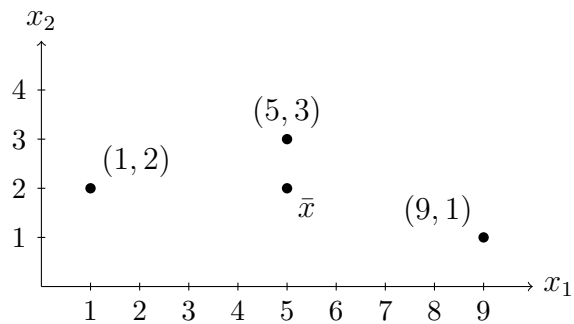
$$(9, 1), \quad (5, 3), \quad (1, 2).$$

The sample mean is

$$\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} (9 + 5 + 1)/3 \\ (1 + 3 + 2)/3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

So the sample mean is

$$\boxed{\bar{x} = (5, 2)}.$$



(b)

The two columns of  $X$  are

$$y_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix}, \quad y_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}.$$

Since

$$\bar{x}_1 = 5, \quad \bar{x}_2 = 2,$$

we have

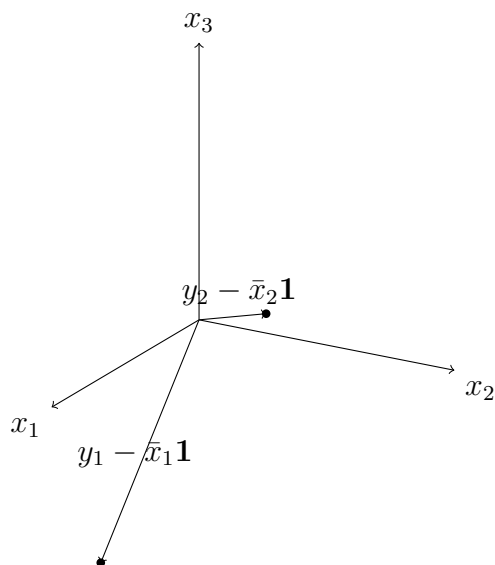
$$\bar{x}_1 \mathbf{1} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad \bar{x}_2 \mathbf{1} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}.$$

Thus,

$$y_1 - \bar{x}_1 \mathbf{1} = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \boxed{\begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}}$$

and

$$y_2 - \bar{x}_2 \mathbf{1} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}.$$



(c)

Let

$$d_1 = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}, \quad d_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

The lengths are

$$\begin{aligned} \|d_1\| &= \sqrt{4^2 + 0^2 + (-4)^2} \\ &= \sqrt{32} = 4\sqrt{2}, \end{aligned}$$

so

$$\boxed{\|d_1\| = 4\sqrt{2}}.$$

Also,

$$\begin{aligned} \|d_2\| &= \sqrt{(-1)^2 + 1^2 + 0^2} \\ &= \sqrt{2}. \end{aligned}$$

Therefore

$$\boxed{\|d_2\| = \sqrt{2}}.$$

The dot product is

$$d'_1 d_2 = 4(-1) + 0(1) + (-4)(0) = -4.$$

Using

$$\cos \theta = \frac{d'_1 d_2}{\|d_1\| \|d_2\|},$$

we get

$$\cos \theta = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = -\frac{1}{2}.$$

Thus

$$\boxed{\cos \theta = -\frac{1}{2}}$$

and

$$\boxed{\theta = 120^\circ}.$$

The mean-corrected matrix is

$$D = X - \mathbf{1}\bar{x}' = \begin{bmatrix} 4 & -1 \\ 0 & 1 \\ -4 & 0 \end{bmatrix}.$$

Using

$$S_n = \frac{1}{n} D' D,$$

we get

$$S_n = \frac{1}{3} \begin{bmatrix} 32 & -4 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 32/3 & -4/3 \\ -4/3 & 2/3 \end{bmatrix}.$$

The correlation matrix is

$$R = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}.$$

The lengths of the deviation vectors and the cosine of their angle are related to the diagonal and off-diagonal entries of  $S_n$ .

## Problem 3.6

Given

$$X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}.$$

The means are

$$\bar{x}_1 = \frac{-1 + 2 + 5}{3} = 2,$$

$$\bar{x}_2 = \frac{3 + 4 + 2}{3} = 3,$$

$$\bar{x}_3 = \frac{-2 + 2 + 3}{3} = 1.$$

Therefore,

$$\bar{x}' = [2 \quad 3 \quad 1].$$

(a)

The deviation matrix is

$$\begin{aligned} D &= X - \mathbf{1}\bar{x}' \\ &= \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}}. \end{aligned}$$

The rows add up to zero, so they are linearly dependent. Therefore,

$$\text{rank}(D) < 3.$$

The first two rows are independent, so

$$\boxed{\text{rank}(D) = 2}.$$

Hence  $D$  is not full rank.

**(b)**

Since  $n = 3$ ,

$$S = \frac{1}{n-1} D' D = \frac{1}{2} D' D.$$

First,

$$D' D = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix}.$$

Therefore,

$$S = \begin{bmatrix} 9 & -\frac{3}{2} & \frac{15}{2} \\ -\frac{3}{2} & 1 & -\frac{1}{2} \\ \frac{15}{2} & -\frac{1}{2} & 7 \end{bmatrix}.$$

Since  $D$  does not have full rank,  $S$  is singular. Thus,

$$|S| = 0.$$

Geometrically, this means the data do not form a three-dimensional volume. They lie in a lower-dimensional space.

**(c)**

The total sample variance is

$$\text{tr}(S) = 9 + 1 + 7 = \boxed{17}.$$

## Problem 3.9

Given

$$X = \begin{bmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{bmatrix}.$$

The means are

$$\bar{x}_1 = \frac{12 + 18 + 14 + 20 + 16}{5} = 16,$$

$$\bar{x}_2 = \frac{17 + 20 + 16 + 18 + 19}{5} = 18,$$

$$\bar{x}_3 = \frac{29 + 38 + 30 + 38 + 35}{5} = 34.$$

Thus,

$$\bar{x}' = [16 \quad 18 \quad 34].$$

(a)

The mean-corrected matrix is

$$D = X - \mathbf{1}\bar{x}'$$

$$D = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}.$$

Let the columns be  $D_1, D_2, D_3$ . We can see that

$$D_3 = D_1 + D_2.$$

Therefore,

$$D_1 + D_2 - D_3 = 0.$$

So one vector that gives the linear dependence is

$$a = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

Indeed,

$$Da = 0.$$

(b)

The sample covariance matrix is

$$S = \frac{1}{n-1} D' D = \frac{1}{4} D' D.$$

We get

$$D' D = \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix}.$$

Therefore,

$$S = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 5/2 & 11/2 \\ 13 & 11/2 & 37/2 \end{bmatrix}.$$

Since the columns are linearly dependent,

$$|S| = 0.$$

Now,

$$Sa = \begin{bmatrix} 10 & 3 & 13 \\ 3 & 5/2 & 11/2 \\ 13 & 11/2 & 37/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

This gives

$$Sa = \begin{bmatrix} 10 + 3 - 13 \\ 3 + \frac{5}{2} - \frac{11}{2} \\ 13 + \frac{11}{2} - \frac{37}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus,

$$Sa = 0.$$

So  $a$  is an eigenvector corresponding to the eigenvalue

$$\lambda = 0.$$

**(c)**

From the original data,

$$29 = 12 + 17,$$

$$38 = 18 + 20,$$

$$30 = 14 + 16,$$

$$38 = 20 + 18,$$

$$35 = 16 + 19.$$

Therefore,

$$\boxed{x_3 = x_1 + x_2}.$$

So the linear dependence is

$$\boxed{a_1 = 1, \quad a_2 = 1, \quad a_3 = -1}.$$

## Problem 3.18

The given mean is

$$\bar{x} = \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix}$$

and the covariance matrix is

$$S = \begin{bmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.128 & 0.067 \\ 0.173 & 0.127 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{bmatrix}.$$

For the calculations, I used  $S_{32} = 0.128$  so that the covariance matrix is symmetric.

(a)

Let

$$T = x_1 + x_2 + x_3 + x_4.$$

Then

$$a = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

The mean is

$$\begin{aligned} \bar{T} &= a'\bar{x} \\ &= 0.766 + 0.508 + 0.438 + 0.161 \\ &= \boxed{1.873}. \end{aligned}$$

The variance is

$$\text{Var}(T) = a'Sa.$$

Therefore,

$$\begin{aligned} \text{Var}(T) &= 0.856 + 0.568 + 0.171 + 0.043 \\ &\quad + 2(0.635 + 0.173 + 0.096 \\ &\quad \quad + 0.128 + 0.067 + 0.039) \\ &= \boxed{3.914}. \end{aligned}$$

Thus,

$$\boxed{\bar{T} = 1.873, \quad s_T^2 = 3.914}.$$

(b)

Let

$$D = x_1 - x_2.$$

Then

$$a = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}.$$

The mean is

$$\bar{D} = 0.766 - 0.508 = \boxed{0.258}.$$

The variance is

$$\begin{aligned} \text{Var}(D) &= S_{11} + S_{22} - 2S_{12} \\ &= 0.856 + 0.568 - 2(0.635) \\ &= \boxed{0.154}. \end{aligned}$$

Therefore,

$$\boxed{\bar{D} = 0.258, \quad s_D^2 = 0.154}.$$

For the covariance with the total variable,

$$T = x_1 + x_2 + x_3 + x_4.$$

Using

$$\text{Cov}(a'X, b'X) = a'Sb,$$

we get

$$\begin{aligned} \text{Cov}(D, T) &= (0.856 + 0.635 + 0.173 + 0.096) \\ &\quad - (0.635 + 0.568 + 0.128 + 0.067) \\ &= 1.760 - 1.398 \\ &= \boxed{0.362}. \end{aligned}$$

Thus,

$$\boxed{\text{Cov}(D, T) = 0.362}.$$

## Problem 3.20

There are 25 observations. Let

$$D = x_2 - x_1.$$

(a)

First calculate the means:

$$\bar{x}_1 = \frac{235.5}{25} = \boxed{9.42},$$

$$\bar{x}_2 = \frac{481.8}{25} = \boxed{19.272}.$$

The sample variances are

$$s_1^2 = \frac{339.34}{24} = \boxed{14.1392},$$

and

$$s_2^2 = \frac{1493.7304}{24} = \boxed{62.2388}.$$

The sample covariance is

$$s_{12} = \frac{323.344}{24} = \boxed{13.4727}.$$

The mean of the difference is

$$\begin{aligned}\bar{D} &= \bar{x}_2 - \bar{x}_1 \\ &= 19.272 - 9.42 \\ &= \boxed{9.852}.\end{aligned}$$

For the variance,

$$s_D^2 = s_2^2 + s_1^2 - 2s_{12}.$$

Thus,

$$\begin{aligned}s_D^2 &= 62.2388 + 14.1392 - 2(13.4727) \\ &= \boxed{49.4326}.\end{aligned}$$

Therefore,

$$\boxed{\bar{D} = 9.852, \quad s_D^2 = 49.4326}.$$

(b)

Now calculate each difference  $d_j = x_{2j} - x_{1j}$ :

$$\begin{array}{ccccc} 1.2, & 2.0, & 9.4, & 2.0, & 4.1, \\ 5.2, & 23.1, & 5.3, & 4.8, & 15.4, \\ 11.7, & 11.5, & 22.5, & 14.2, & 8.8, \\ 7.1, & 5.5, & 4.8, & 22.6, & 6.5, \\ 24.8, & 7.0, & 9.8, & 4.4, & 12.6. \end{array}$$

Their sum is

$$\sum d_j = 246.3.$$

Therefore,

$$\bar{d} = \frac{246.3}{25} = \boxed{9.852}.$$

Also,

$$\sum d_j^2 = 3612.93.$$

Using

$$s_d^2 = \frac{\sum d_j^2 - \frac{(\sum d_j)^2}{n}}{n - 1},$$

we get

$$\begin{aligned} s_d^2 &= \frac{3612.93 - \frac{(246.3)^2}{25}}{24} \\ &= \boxed{49.4326}. \end{aligned}$$

Therefore,

$$\boxed{\bar{d} = 9.852, \quad s_d^2 = 49.4326}.$$

The results are the same as in part (a).